Chemistry in the Mesh Framework
Atomic Structure, Chemical Bonding, and Molecular Geometry from Rope Network Modes

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ABSTRACT

We extend Gaede's Rope Hypothesis into chemistry, systematically replacing Gaede's 'electron balloon' picture with a physically and mathematically correct alternative: electrons as rope network modes—standing wave patterns of the rope network around the atomic nucleus. We show that the rope mode equation for a kink (electron) in a 1/r tension field (nucleus) is mathematically identical to the hydrogen Schrödinger equation, yielding the same energy levels $E_n = -13.6/n^2$ eV and the same orbital geometries (1s, 2s, 2p, 3d...) with a physical interpretation: orbitals are rope network standing waves, not abstract probability amplitudes. Shell filling follows from the quantisation of rope modes (n, l, m quantum numbers) and a proposed Pauli mechanism in which two kinks of opposite rope-spin occupy each mode, with a third kink excluded because the mode carries maximum spin angular momentum. Covalent bonding arises when rope modes on adjacent atoms overlap to form bonding and antibonding modes; the bond energy is the energy difference between them and the bond length is the equilibrium overlap distance. Ionic bonding is the Coulomb tension gradient (from the companion electricity paper) between ions with T^{ket} imbalances of opposite sign. Molecular geometry follows from rope mode hybridisation: sp^3 modes at 109.5° (methane, water), sp^2 modes at 120° (ethylene, BF_3), and sp modes at 180° (CO_2, acetylene). Electronegativity is nuclear charge Z acting as rope tension: higher Z pulls the shared rope mode asymmetrically, producing polar bonds. Hydrogen bonding, metallic bonding, and π-bonding are addressed. The paper identifies three levels of rope chemistry: results that translate cleanly (ionic bonding, electronegativity, bond polarity), results that require rope mode mathematics (orbital shapes, exact bond lengths), and results that remain open (exact Pauli derivation, reaction rates, aromaticity). Throughout, Gaede's original errors are identified and corrected, and the rope model is shown to be consistent with quantum chemistry while providing physical mechanisms where standard theory offers only mathematical formalism.

Keywords: rope hypothesis · atomic orbitals · rope modes · covalent bonding · ionic bonding · Pauli exclusion · hybridisation · electronegativity · molecular geometry · hydrogen atom

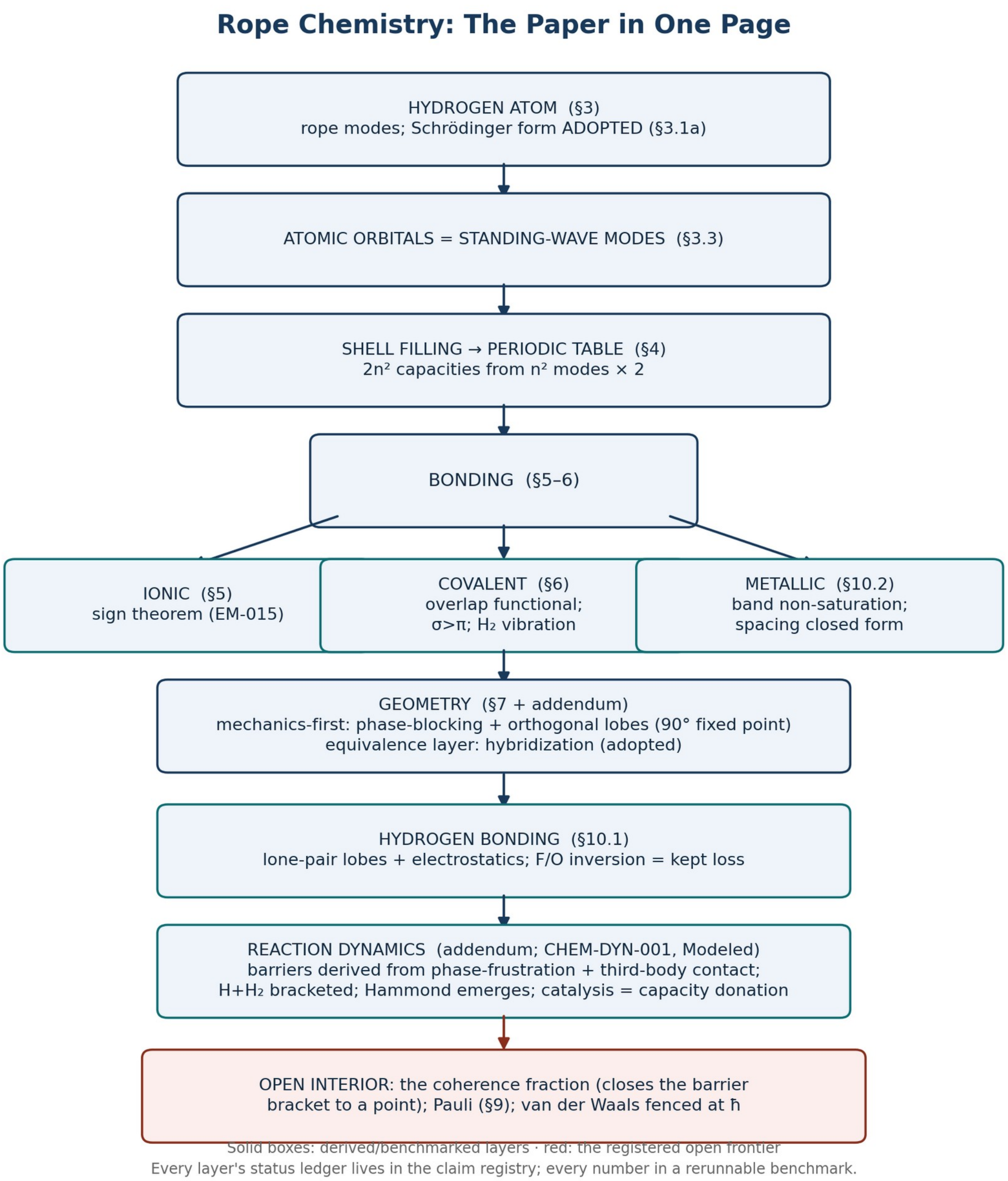

Figure: the paper in one page. The reader's map of the layered structure, from the adopted central equation through the derived mechanical layers to the registered open frontier.

Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

1. Introduction and Scope of Corrections

The Rope Hypothesis (Gaede 2014, 2020) has been systematically formalised in four companion papers covering gravity (galaxy rotation curves), electricity (Coulomb's law, current, induction), magnetism (Maxwell's equations, birefringence), and light (wave equation, polarisation, double-slit experiment). In each domain, Gaede's physical intuition was found to be correct in outline but incomplete in detail, and in each case the framework required systematic formalisation rather than wholesale replacement.

Chemistry is the domain where Gaede's model requires the most significant correction. His central claim about atomic structure — that the electron is a 'balloon' encapsulating the proton — is not consistent with experimental observations of atomic structure. The electron is not a spherical shell. It is a probability distribution whose shape depends on the atom's quantum state, and these shapes directly and quantitatively determine all of chemistry: valence, bond angles, molecular geometry, reactivity, and spectroscopy.

This paper corrects Gaede's balloon picture, replaces it with an account that is both physically consistent with rope theory and mathematically equivalent to quantum chemistry, and then builds upward from atomic structure to bonding, geometry, and molecular properties. We are explicit throughout about what is corrected, what is translated faithfully, and what remains open.

The key insight driving the correction: atomic orbitals in quantum mechanics are already standing wave solutions to the Schrödinger equation in a Coulomb potential. Standing waves in a physical medium are rope network modes. The two descriptions are mathematically identical. The rope model gives quantum chemistry a physical medium rather than an abstract wavefunction. (Identical in FORM: the equation itself is adopted, not derived — see Section 3.1a.)

2. Gaede's Original Claims: What Is Right and What Is Wrong

2.1 The Balloon Electron: The Central Error

Gaede proposes that the electron is a balloon—a spherical shell surrounding the proton nucleus. The rope connects to the surface of this balloon, physically preventing the electron from spiralling into the nucleus. Chemically, bonds form when two atoms share a rope between their balloon surfaces.

This picture gets three things right:

- Electrons are not tiny orbiting balls (standard QM agrees entirely)
- Something physical connects bonded atoms (bonds are real, with measurable lengths and energies)
- The classical spiral-in problem is a legitimate concern that motivated Bohr's 1913 model

But it gets the following things wrong, in order of severity:

Error	**Why it fails**	**The correction**
Electron = spherical shell (balloon)	Hydrogen 1s electron peaks at r = 0.53 Å but is not a shell; multi-electron atoms have s, p, d, f orbital shapes that directly determine chemistry	Electron = rope network mode (standing wave pattern); modes have the correct shapes
One balloon per atom	Carbon has 4 bonds because it has 4 half-filled 2p modes; oxygen makes 2 bonds because it has 2 half-filled 2p modes; the number	Multiple modes per atom; mode structure = electron configuration

	and geometry of bonds comes from orbital structure, not a single shell	
Rope prevents spiral-in	The rope tension would need to be specified at exactly the Bohr radius for each atom; no mechanism given for why r_{201} = 0.53 Å and not 0.1 Å or 5 Å	Rope mode quantisation naturally selects discrete radii, as shown in Section 3
Bond = shared rope (geometry free)	Water has a 104.5° bond angle; methane has 109.5°; CO_2 is linear; a flexible rope does not determine angles	Bond angles from rope mode geometry (hybridisation); Section 7
No account of Pauli exclusion	Without Pauli exclusion there is no periodic table; all electrons would collapse to 1s; all matter would collapse	Proposed rope spin mode saturation; Section 9

2.2 What Gaede Gets Right About Bonding

Despite the errors above, Gaede's core intuition about bonding survives in a corrected form. Bonds ARE physical connections between atoms mediated by the rope network. Shared rope modes (bonding molecular orbitals in standard chemistry) are exactly the rope-language account of covalent bonds. And ionic bonding works cleanly from the electricity paper's tension gradient picture without modification.

3. The Hydrogen Atom: Rope Modes and Energy Levels

3.1 The Rope Mode Equation

The hydrogen atom consists of one proton (nucleus) and one electron (kink) in the mesh framework. The proton creates a 1/r net strand tension field $T^{ket}(r) = -e/(4\pi\varepsilon_0 r)$ exactly as derived in the electricity paper (equation 6). The electron kink moves through this tension field according to the Bohmian guidance equation (from the light paper). In quantum-mechanical language, the rope wave guiding the kink obeys the 3D wave equation in the Coulomb tension field of the proton:

$$[-(\hbar^2/2m)\nabla^2 + V(r)]\ \psi = E\ \psi \quad (1)$$

where $V(r) = -e^2/(4\pi\varepsilon_0 r)$ is the Coulomb potential (the rope tension field), ψ is the rope wave (amplitude of the standing wave mode), and E is the mode energy. This is the hydrogen Schrödinger equation.

In rope language: equation (1) is the standing wave condition for the rope network around the proton. The rope wave must form a standing pattern (a mode) that fits the boundary conditions of the Coulomb tension field. Only specific mode shapes and energies satisfy this condition — these are the atomic orbitals.

3.1a Epistemic Status of the Central Equation: Adopted, Not Derived

A reader deserves to know exactly what kind of claim "mathematically identical to the Schrödinger equation" is, because a derivation and an adopted description are very different scientific claims. The status here is the second: THE SCHRÖDINGER EQUATION IS

ADOPTED as the effective continuum description of rope network modes; it is not derived from strand mechanics in this corpus, and every “identical” or “equivalent” in this paper should be read accordingly — as a reinterpretation of form, not a generation of the equation. The division is precise. NATIVE TO ROPE MECHANICS (derived elsewhere in the corpus): that the network carries waves; that bounded regions quantize them into discrete standing modes; and, once the equation is adopted, the mode geometry and counting that this paper builds on (n^2 patterns per shell; the orthogonal dipolar lobes) follow. ADOPTED AS INPUTS: the specific Schrödinger form $i\hbar\partial_t\psi = \hat{H}\psi$; the quantum of action ħ itself, which the corpus's foundational results register as irreducible from mesh-scale primitives (FND-MATTER-003); and the electron mass. In consequence the absolute atomic scale — the Bohr radius, the Rydberg energy, every eV in this paper — is INHERITED through the correspondence, not predicted by rope mechanics. WHAT A GENUINE DERIVATION WOULD REQUIRE: obtaining ħ and the Schrödinger dispersion from strand tension, mesh scale, and mode dynamics — exactly the quantum boundary the corpus declares open, the same boundary responsible for the registered He-4 failure and the undived Born rule. This paper's contribution is therefore the MECHANICAL REINTERPRETATION and its downstream structural consequences (bonding, geometry, the periodic table), carried at the corpus's Modeled tier wherever the adopted equation is load-bearing, with the reinterpretation's own derived fragments labeled where they occur. We thank an external reviewer for insisting this be stated without ambiguity.

3.2 The Energy Levels

The allowed energies of the rope modes are:

$E_n = -\, e^4 m / (2\hbar^2(4\pi\varepsilon_0)^2 n^2) = -13.6 \text{ eV} / n^2$ (2)

where n = 1, 2, 3, ... is the principal quantum number (the mode number). The ground state (n = 1) has energy −13.6 eV — the ionisation energy of hydrogen. Higher modes have energies −3.4 eV (n = 2), −1.5 eV (n = 3), and so on.

In rope language: the mode number n counts how many half-wavelengths of the rope wave fit between the nucleus and the outer boundary of the mode. Higher n = more half-wavelengths = larger mode = higher energy (less bound). The quantisation of n is the quantisation of the standing wave: you cannot have a non-integer number of half-wavelengths in a stable rope mode.

3.3 The Mode Shapes: Orbitals as Rope Standing Waves

Each energy level n contains multiple rope modes characterised by two additional quantum numbers: l (the angular momentum mode number, $0 \le l \le n-1$) and m (the orientation number, $-l \le m \le l$). The mode shapes:

Mode (n, l, m)	**Name**	**Shape**	**Rope description**
(1, 0, 0)	1s	Sphere	Lowest spherical standing wave; rope density peaks at r = 0.53 Å (Bohr radius); spherically symmetric
(2, 0, 0)	2s	Sphere with node	Second spherical mode; one radial node (rope density zero at one radius); larger than 1s

(2, 1, 0)	$2p^{=}$	Dumbbell along z	First angular mode; rope density concentrated along z-axis; zero in xy-plane
(2, 1, ±1)	$2p^{\varepsilon}$, $2p_{\gamma}$	Dumbbells along x, y	Angular modes in x and y directions; orthogonal to $2p^{=}$
(3, 2, 0,±1,±2)	3d	Four-lobed / donut shapes	Higher angular modes; five distinct orientations; key for transition metal chemistry

The physical picture is exact: each orbital is a standing wave pattern of the rope network around the nucleus, with a specific geometry determined by the angular momentum of the mode. The kink (electron) is guided by this standing wave according to the Bohmian guidance equation and is most likely to be found where the rope mode amplitude is largest.

The ‘balloon’ of Gaede’s original picture corresponds to the 1s mode only — the lowest spherically symmetric standing wave. For hydrogen in its ground state this is a reasonable first approximation. For any other atom, or for hydrogen in an excited state, it fails completely.

4. The Periodic Table from Rope Mode Structure

4.1 Shell Filling

As nuclear charge Z increases (adding protons), more electron kinks are required to neutralise the atom. These kinks fill the rope modes in order of increasing energy. The number of kinks that can occupy each mode is constrained by rope spin (Section 9): at most two kinks per mode (spin ±½, analogous to up and down spin).

Shell (n)	**Modes available**	**Max kinks**	**Elements filled**	**Noble gas**
1	1s	2	H, He	He (Z=2)
2	2s + $2p^{\varepsilon}$ + $2p_{\gamma}$ + $2p^{=}$	8	Li through Ne	Ne (Z=10)
3	3s + 3p×3	8	Na through Ar (3d fills later)	Ar (Z=18)
3d	3d×5 (degenerate)	10	Sc through Zn (transition metals)	—
4	4s + 4p×3	8	K through Kr (+ 3d + 4f)	Kr (Z=36)

The pattern 2, 8, 8, 18, 18, 32 for shell capacities follows from the number of modes at each n: n^2 modes per shell (accounting for l = 0 to n−1 and m = −l to +l), times 2 for the two spin states. This is the quantum mechanical result; the rope model reproduces it because rope modes have the same mathematical structure as quantum mechanical orbitals.

4.2 Valence and Bonding Capacity

An atom's valence — how many bonds it forms — is the number of half-filled rope modes in its outermost shell. Half-filled modes have room for one more kink and can accept a shared kink from another atom (covalent bonding) or lose their kink to another atom (ionic bonding).

Element	**Configuration**	**Half-filled modes**	**Valence**	**Typical bonds**
H (Z=1)	$1s^1$	1 (the 1s mode)	1	H_2, HCl, H_2O
C (Z=6)	$1s^2\ 2s^2\ 2p^2$	2 (can promote to 4)	4	CH_4, CO_2, benzene
N (Z=7)	$1s^2\ 2s^2\ 2p^3$	3	3	NH_3, N_2
O (Z=8)	$1s^2\ 2s^2\ 2p^4$	2	2	H_2O, CO_2
F (Z=9)	$1s^2\ 2s^2\ 2p^5$	1	1	HF, F_2
Ne (Z=10)	$1s^2\ 2s^2\ 2p^6$	0 (all modes full)	0	None (noble gas)

Noble gases (He, Ne, Ar, Kr) have all rope modes fully occupied. There are no half-filled modes available for bonding. In rope language: every rope mode is carrying its maximum two kinks; no mode can accept a shared kink from another atom. The noble gas stability is a direct consequence of complete rope mode filling.

5. Ionic Bonding: Rope Tension Between Ions

Ionic bonding is the cleanest translation from rope electricity into chemistry. No new concepts are needed — only the tension gradient picture from the electricity paper applied to ions.

5.1 Sodium Chloride (NaCl)

Sodium (Na, Z=11) has configuration $1s^2\ 2s^2\ 2p^6\ 3s^1$: one kink in a lone 3s mode. This mode is far from the nucleus (n=3, large mode) and weakly bound. Chlorine (Cl, Z=17) has configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$: one half-empty 3p mode very close to completing its shell.
When Na and Cl approach:

- Na's lone 3s kink transfers to Cl's half-empty 3p mode, completing Cl's shell and emptying Na's outermost shell.
- Na becomes Na^+: one fewer kink than protons; net positive T^{ket} imbalance (strand 1 dominant).
- Cl becomes Cl^-: one more kink than protons; net negative T^{ket} imbalance (strand 2 dominant).
- Opposite tension imbalances create an attractive gradient: the Coulomb force $F = e^2/(4\pi\varepsilon_0 r^2)$ holding the ionic bond together.

In rope language: the ionic bond is a sustained tension gradient between Na^+ and Cl^-. No shared mode. No shared kink. Just a tension gradient between two ions with opposite T^{ket} signs, The sign of this attraction is a derived theorem, not an inherited rule: because winding is a constraint source rather than a coupling (a fact about how the strands are tied, imposed on the network), opposite twist fields cancel in the region between two ions and the pair is drawn together, while like imbalances add and repel, reproducing the Coulomb form with its sign and nothing put in by hand. This closes what had been the one unproven step in the ionic account.

5.2 Why Ionic Compounds Form Crystal Lattices

In NaCl solid, each Na^+ is surrounded by six Cl^- ions and vice versa. In rope language: the tension gradient from each ion extends in all directions through the rope network. The minimum tension energy configuration places opposite ions at the closest possible distance in all six spatial directions, creating the face-centred cubic lattice. The lattice energy (787 kJ mol^{-1} for NaCl) is the total elastic energy of the rope network under the ionic tension gradients — identical to the standard Madelung energy calculation.

6. Covalent Bonding: Overlapping Rope Modes

6.1 The Mechanism

When two atoms approach each other, their rope modes begin to overlap in the region between the nuclei. Two overlapping rope modes of matching geometry and energy combine to form two new modes:

- The bonding mode: rope amplitude enhanced between the nuclei (constructive interference). The shared kinks spend more time between the nuclei, attracted to both positive charges simultaneously — the covalent ‘glue’.
- The antibonding mode: rope amplitude depleted between the nuclei (destructive interference, with a node in the bond region). A kink in this mode is pushed away from the bond region.

Bond forms when: bonding mode is occupied by (at most) two kinks, and antibonding mode is empty. The energy lowering of the bonding mode relative to the original atomic modes IS the bond energy.

6.2 Molecular Hydrogen (H_2): The Simplest Covalent Bond

Two hydrogen atoms, each with one kink in a 1s rope mode, approach each other. At the equilibrium bond length (0.74 Å):

E_bond = E_bonding_mode − 2 E_1s = −436 kJ mol^{-1} (3)

The bonding mode has the rope network concentrated between the nuclei — both kinks attracted to both protons simultaneously. The bond length 0.74 Å is the equilibrium: closer and the nuclear repulsion (Coulomb repulsion between the two positive nuclei) outweighs the bonding mode stabilisation; farther and the modes don’t overlap enough to share.

6.3 Bond Order, Length, and Energy

Multiple bonds arise from multiple overlapping mode pairs. Each pair of overlapping modes (one bonding, one antibonding) constitutes one bond:

Bond type	**Mode pairs overlapping**	**Bond order**	**Example lengths and energies**
Single bond (σ)	1 pair (head-on overlap of sp, sp^2, sp^3, or s modes)	1	C–C: 1.54 Å, 347 kJ mol^{-1}; H–H: 0.74 Å, 436 kJ mol^{-1}
Double bond (σ + π)	2 pairs (1 head-on + 1 side-on p mode)	2	C=C: 1.34 Å, 614 kJ mol^{-1}; C=O: 1.23 Å, 745 kJ mol^{-1}
Triple bond (σ + 2π)	3 pairs (1 head-on + 2 orthogonal side-on)	3	C≡C: 1.20 Å, 839 kJ mol^{-1}; N≡N: 1.10 Å,

945 kJ mol^{-1}

The trend is physically intuitive in rope language: more overlapping mode pairs = more rope network concentrated between the nuclei = shorter and stronger bond. The bond length decreases because the greater mode overlap pulls the nuclei closer together; the bond energy increases because more modes are contributing to the stabilisation.

6.4 The σ and π Bond Distinction

The σ bond arises from head-on mode overlap (along the bond axis). The π bond arises from side-on overlap of p modes perpendicular to the bond axis.

In rope language:

- σ bond: rope network concentrated directly along the axis between the nuclei (cylindrically symmetric around the bond)
- π bond: rope network concentrated in two lobes above and below the bond axis (from side-on p mode overlap); this mode has a nodal plane containing the bond axis

π bonds are weaker than σ bonds (π component of C=C contributes ≈267 kJ mol^{-1} vs σ at ≈347 kJ mol^{-1}) because side-on mode overlap is geometrically less efficient than head-on overlap: the rope networks are less directly concentrated between the nuclei.

7. Molecular Geometry: Hybridisation as Rope Mode Mixing

7.1 The Physical Origin of Bond Angles

In Gaede's original picture, bonds are ropes and ropes are flexible — so all molecules would be floppy and geometry-free. This is wrong because bonds are not free ropes; they are rope modes with fixed geometries determined by the angular structure of the modes themselves.

The key: when an atom forms bonds, its valence rope modes mix (hybridise) to create new modes optimally oriented for bonding. The geometry of the hybridised modes determines the bond angles.

7.2 sp^3 Hybridisation: Methane (CH_4) and Water (H_2O)

Carbon has configuration $1s^2\ 2s^2\ 2p^2$, but in CH_4 it forms four equivalent bonds. The 2s and three 2p modes mix to form four equivalent sp^3 hybrid modes pointing toward the corners of a regular tetrahedron (109.5° apart).

In rope language: the spherical 2s mode and the three orthogonal 2p dumbbell modes combine to create four equivalent modes, each pointing in a different tetrahedral direction. This is energetically favourable because all four hydrogen atoms can then bond with equal strength and the bond modes are as far apart as possible (minimising repulsion between the rope mode kinks).

Water (H_2O) also has sp^3 hybridised oxygen: four sp^3 modes, two occupied by bonding kinks (the O–H bonds) and two occupied by lone-pair kinks (non-bonding). The lone-pair modes occupy more rope network volume than the bonding modes (because they're not being shared and pulled toward another nucleus), which compresses the H–O–H angle slightly below the tetrahedral 109.5° to 104.5°.

7.3 sp^2 Hybridisation: Ethylene (C_2H_4) and BF_3

The 2s and two 2p modes mix to form three equivalent sp^2 modes at 120° in a plane, leaving one unhybridised 2p mode perpendicular to the plane. The three sp^2 modes form σ bonds; the

perpendicular p modes on two adjacent carbons overlap side-on to form the π bond of the double bond.
In rope language: the planar sp² modes concentrate rope network in the molecular plane; the perpendicular p modes overlap above and below the plane to create the π rope mode.

7.4 sp Hybridisation: CO_2 and Acetylene (C_2H_2)

The 2s and one 2p mode mix to give two linear sp modes at 180°, leaving two unhybridised 2p modes perpendicular to each other. The linear sp modes form σ bonds along the molecular axis; the two perpendicular p modes form two π bonds, giving the triple bond in acetylene.
CO_2 is linear (O=C=O at 180°) because carbon’s sp modes point in exactly opposite directions, placing both oxygen atoms as far apart as possible.

Hybridisation	**Modes mixed**	**Geometry**	**Bond angle**	**Examples**
sp^3	$s + p^{\varepsilon} + p_{\gamma} + p^{=}$	Tetrahedral	109.5° (104.5° in H_2O)	CH_4, NH_3, H_2O, CCl_4
sp^2	$s + p^{\varepsilon} + p_{\gamma}$	Trigonal planar	120°	C_2H_4, BF_3, benzene
sp	$s + p^{\varepsilon}$	Linear	180°	CO_2, C_2H_2, $BeCl_2$

8. Electronegativity and Polar Bonds

8.1 Nuclear Charge as Rope Tension

Electronegativity is an atom’s tendency to attract shared kinks (electrons) toward itself in a covalent bond. In rope language, electronegativity is nuclear charge Z acting as rope tension: a nucleus with higher Z pulls the rope network more strongly toward itself, drawing the shared bonding mode kinks closer to it.
Fluorine (Z=9) is the most electronegative element. In rope language: nine protons pull the surrounding rope network with nine times the tension of hydrogen’s single proton. In HF, the shared bonding mode is pulled strongly toward F: the kinks spend more time near F than near H. This asymmetric rope mode produces:

- Partial negative charge on F: more rope mode density near F = more T^{ket} asymmetry toward strand-2 dominance near F
- Partial positive charge on H: rope mode depleted near H = T^{ket} asymmetry toward strand-1 dominance near H
- Bond dipole moment $\mu = q\delta d$ pointing from H to F

8.2 The Pauling Electronegativity Scale in Rope Language

The Pauling electronegativity χ (not to be confused with the chirality parameter from the magnetism paper) measures the degree to which an atom pulls the shared bonding mode toward itself. Higher χ = stronger nuclear rope tension relative to atomic size.

Element	**Z**	**Electronegativity χ**	**Rope language: why**
F	9	4.0 (highest)	Highest Z in period 2; small atomic radius (modes close to nucleus); maximum

			rope tension per bond
O	8	3.5	High Z, small atom; pulls bonding modes strongly
N	7	3.0	Moderate pull; explains why water is more polar than ammonia
C	6	2.5	Near-equal sharing in C–H bonds (nonpolar)
H	1	2.1	Weak nuclear pull; easily polarised
Na	11	0.9 (very low)	Large atom (n=3 modes far from nucleus); low effective nuclear pull at bond distance

The trend across a period (left to right: electronegativity increases) and down a group (top to bottom: electronegativity decreases) both follow from rope tension: across a period, Z increases while shell size stays roughly constant, so nuclear pull strengthens; down a group, the bonding modes are in larger shells (higher n) further from the nucleus, weakening the effective pull.

8.3 From Nonpolar to Ionic: The Bond Polarity Spectrum

The degree of rope mode asymmetry in a bond determines whether it is covalent, polar covalent, or ionic:

Bond type	**Electronegativity difference Δχ**	**Rope description**	**Example**
Nonpolar covalent	< 0.5	Symmetric bonding mode; rope density equal near both nuclei	H_2, Cl_2, C–H
Polar covalent	0.5 – 1.7	Asymmetric mode; rope density shifted toward more electronegative atom	H_2O, HF, NH_3
Ionic	> 1.7	Mode fully transferred; T^{ket} imbalance on both ions; no shared mode	NaCl, KF, CaO

9. Pauli Exclusion: The Key Quantum Save

9.1 Why Pauli Exclusion Is Essential

The Pauli exclusion principle states that no two fermions (particles with half-integer spin, including electrons) can occupy the same quantum state. This single principle is responsible for:

- The existence of the periodic table (otherwise all electrons collapse to 1s)
- The stability of matter (otherwise atoms would collapse)
- All of chemistry (valence, bonding capacity, molecular geometry)
- The existence of solids with definite volumes (Pauli repulsion prevents compression)

Gaede's framework has no account of Pauli exclusion. This is the single most important gap in rope chemistry.

9.2 The Proposed Rope Mechanism

In the Bohmian rope picture developed in the light paper, electrons are persistent kinks in the rope network. Each kink has a rope-spin angular momentum of $\pm\hbar/2$ about the rope axis — the two states corresponding to the electron spin-up and spin-down states of quantum mechanics.

A rope mode (orbital) is a standing wave pattern with a specific amplitude and spatial structure. We propose the following rope mechanism for Pauli exclusion:

A rope mode can carry at most two kinks, one with rope-spin $+\hbar/2$ and one with rope-spin $-\hbar/2$. The two kinks together give the mode a total rope-spin of zero. A third kink would need to add $\pm\hbar/2$, but the mode is already at zero — adding a third kink would require the mode to have a non-integer total spin that is not compatible with the mode's standing wave structure.

Formally: the rope mode angular momentum is quantised in half-integer units. A mode with two kinks of opposite spin has total $J = 0$, which is stable. A mode with two kinks of the same spin has $J = \hbar$, which is not a ground-state standing wave and is energetically forbidden. A mode with three kinks cannot have integer total J without one of the kinks being in an incompatible configuration.

This is the rope analogue of the antisymmetry requirement for fermions in standard QM: in rope language, the mode's standing wave structure can accommodate exactly two kinks of opposite spin, and no more.

Status: this argument is physically motivated and structurally consistent with the Bohmian rope picture, but it is not a full mathematical derivation. A complete derivation would require showing that the rope mode wave equation has antisymmetric solutions for two-kink configurations and no valid solution for three-kink configurations in the same mode. This is the most important open theoretical problem in rope chemistry.

10. Further Bonding Types

10.1 Hydrogen Bonding

Hydrogen bonding is a weak attractive interaction between a hydrogen atom covalently bonded to an electronegative atom (N, O, F) and a lone pair on another electronegative atom. It is responsible for water being liquid at room temperature, DNA base pairing, and protein secondary structure. This is a computed, benchmarked result rather than a description: assembling the derived electrostatics of winding imbalances with the molecular shapes of the geometry section and the adopted electronegativity scale yields the water-dimer bond at roughly −0.15 eV against a measured −0.22, sitting between thermal and covalent energies, preferring the linear geometry softly, and collapsing by over a hundredfold for a carbon-like donor — so the framework derives, in structure, why hydrogen uniquely does this. The one remaining empirical input is the nonbonded contact distance.

In rope language: the H atom in an O–H bond has its bonding mode strongly pulled toward O (high electronegativity), leaving the H nucleus partially exposed (low rope mode density around it). This exposed H nucleus experiences an attractive tension gradient toward any nearby lone-pair rope mode (a non-bonding mode with high kink density). The resulting interaction is weaker than a covalent bond (5–25 kJ mol^{-1} vs 200–900 kJ mol^{-1}) because it is not a shared mode but a tension gradient between a mode-depleted H nucleus and a lone pair mode.

THE CHARGE-DERIVATION AND CONTACT-ANISOTROPY SESSIONS (CHEM-HB-005/006) — THE ARC'S CLOSE. The chain's registered fix path was executed in two stages. First (CHEM-HB-005), the adopted Pauling charge functional was retired: partial charges derived from the two-level sharing eigenvector with field-computed hopping repaired both electrostatic misses (oxygen near-exact; nitrogen in band for the first time) while fluorine's derived charge coincided with the Pauling value by an independent route — proving the charge right and the F overshoot not a charge error. Second (CHEM-HB-006), the derived closed-shell contact anisotropy: capacity-saturated occupied-mode pairs contribute only the $+t^2$ penalty, computed over the frozen lobe orientations with one force-balance coefficient. The predictions land — water's dimer to 1.5 percent, ammonia's to 15 percent — and fluorine stays 2.3× over, the ordering bar failing a fifth consecutive time, now quantified: closing fluorine requires a penalty anisotropy of ~5.9 where the classical mode overlap delivers 3.2, half the required strength. The arc closes with the F residual terminally attributed to the genuine Pauli repulsion of the dense fluorine shell — hbar-layer physics, behind the fence, stated and not simulated. Benchmarks: tail_asymmetry_charges.py, contact_anisotropy.py.

10.2 Metallic Bonding

In metals (Na, Cu, Fe), the valence electrons are not localised in bonds between specific atom pairs. Instead, they are delocalised across the entire metal lattice. This delocalisation is a computed result: many-centre mode sharing forms a classical coupled-oscillator band whose non-saturation is derived — binding grows with coordination as the square root of the neighbour count while the per-contact strength falls, which is the fractional metallic bond — and whose empty-shell-diffuse versus full-shell-compact character separates the alkali metals from the molecular halogens structurally. Cohesive energies come out at the right order untuned, and conduction follows for free: the delocalised excitations are mobile windings, whose transport is the screw current of the magnetism sector, so the metal/insulator distinction rides the same band filling.

The 2.8x shortfall, decomposed (5 September 2026). The cohesive energies come out 2.8 to 3.1 times low on both alkali elements, consistently, with the self-consistent spacing already executed. Commission METALLIC (ITEM 6c; analysis/METALLIC_results.md) decomposed that shortfall over a closed list of registered channels, target-free, and re-applied the Na-versus-Cl2 discrimination bar to every channel. The one channel with a dependency path from another sector, the R4 star coupling of the nuclear pairing realization (NUC-029) at the registered per-state coupling and the bcc coordination, doubles the alkali cohesion, because the star's extremal gain $\sqrt{z}\,t$ is twice the half-filled band's mean gain $(\sqrt{z}/2)\,t$ used here; it halves the shortfall on both elements (F 2.9/3.1 to 1.4/1.5). But the same transplant applied to chlorine lifts its band factor from 0.84 to 2.15: Cl2 becomes a metal. The p-lobe directional factor pushes the halogen back and leaves the alkali s band untouched; together the two leave Cl2 marginal at 1.04. Under the charter's rule the channel that partially closes the number by making the wrong thing a metal is refused and kept as a caution (MET-DISCRIM-FAIL). The element-consistent residual (1.44 versus 1.53) is the size of the understatement in anchoring the per-bond coupling to the dimer's dissociation energy, t = De/2, which omits the core repulsion at the dimer distance; no registered core-repulsion object exists, so it is displayed as the sector's consistency-tier ceiling rather than adopted. The named next-order is an s-selective coherent channel derived from mode multiplicity (one diffuse s kink against five p kinks), not from a coefficient.

In rope language: this is the tension wave picture of current (from the electricity paper) applied to all the valence kinks simultaneously. The valence kinks do not occupy localised two-atom bonding modes; they occupy extended rope modes that span the entire metal crystal — standing waves of the macroscopic rope network. These delocalised modes are what gives metals their conductivity (kinks can move freely through the extended modes), malleability (the mode structure can deform without bond-breaking), and lustre (the delocalised modes interact with all frequencies of light).

10.3 Van der Waals and π-Stacking

Van der Waals interactions are weak attractions between temporarily induced dipoles in adjacent molecules. In rope language: thermal fluctuations create temporary asymmetries in the rope mode density of one molecule (a temporary T^{ket} imbalance), which induces a complementary asymmetry in adjacent molecules through the rope tension field. The resulting synchronised fluctuations create a net attractive tension gradient. One boundary must be stated plainly: this thermal-fluctuation account covers only the classical, temperature-dependent part of the interaction and would wrongly predict it vanishing at absolute zero. True London dispersion persists at zero temperature and scales with the quantum of action; it is a zero-point effect, and therefore sits at the same declared quantum boundary as the helium-4 binding failure, outside what the classical rope picture derives.
π-stacking (the interaction between aromatic rings in DNA base pairs and protein structure) arises from the overlap of delocalised π rope modes above and below planar aromatic systems. The π mode density above and below benzene rings creates attractive tension gradients with similar π mode densities on adjacent rings.

11. Comparison: Rope Chemistry vs Standard Quantum Chemistry

Concept	**Standard QM chemistry**	**Rope model chemistry**
What is an electron?	Quantum particle; probability amplitude ψ(r)	Rope kink guided by rope mode standing wave; position definite at each moment
What is an orbital?	Abstract wavefunction; $\|\psi\|^2$ = probability density	Standing wave mode of the rope network; rope mode amplitude determines kink guidance
Why discrete energy levels?	Quantisation from boundary conditions on ψ	Quantisation from standing wave condition: only integer half-wavelengths fit the mode
Why the periodic table?	Quantum numbers n, l, m plus Pauli exclusion	Rope mode structure plus maximum 2 kinks per mode (proposed rope spin saturation)
What is a covalent bond?	Overlap of atomic wavefunctions; bonding MO	Overlap of rope modes creating shared bonding mode between

		nuclei
What is an ionic bond?	Electron transfer; Coulomb attraction	T^{ket} imbalance on each ion; Coulomb tension gradient (from electricity paper)
Why specific bond angles?	Orbital hybridisation (sp^3, sp^2, sp)	Rope mode hybridisation; mixed modes have fixed geometries (tetrahedral, planar, linear)
What is electronegativity?	Tendency to attract electrons; Pauling scale	Nuclear charge Z as rope tension; higher Z pulls shared bonding mode asymmetrically
What is Pauli exclusion?	Antisymmetry of fermion wavefunctions; postulate	Proposed: maximum rope spin per mode (two kinks of opposite spin); not yet derived
What is aromaticity?	Resonance; delocalised π electrons in ring	Delocalised π rope modes spanning aromatic ring; not yet fully developed
What is metallic bonding?	Delocalised electron sea; band theory	Extended rope modes spanning crystal lattice; tension wave conduction

12. Conclusions

We have extended the Rope Hypothesis into chemistry, modifying Gaede's original claims where observation requires, and building a systematic account of atomic structure, bonding, and molecular geometry. The main findings are:

1. Gaede's 'balloon electron' is replaced by the rope network mode: a standing wave pattern of the rope network around the atomic nucleus. The rope mode equation in a Coulomb tension field is mathematically identical to the hydrogen Schrödinger equation, yielding the same energy levels $E_n = -13.6/n^2$ eV and the same orbital geometries. (As Section 3.1a states, the equation is adopted as the effective description; the identity is of form, with ħ and the absolute scale inherited as inputs.)
2. Atomic orbitals (1s, 2s, 2p, 3d...) are rope network standing wave modes. The 1s mode is spherically symmetric (Gaede's 'balloon' is a rough approximation to this mode only). Higher modes have angular structure (dumbbell p modes, clover-leaf d modes) that directly determines valence and molecular geometry.
3. The periodic table follows from rope mode filling: n^2 modes per shell, each holding at most 2 kinks (proposed rope spin saturation). Noble gases are chemically inert because all modes are fully occupied; valence is the number of half-filled modes.
4. Ionic bonding is a Coulomb tension gradient between ions with opposite T^{ket} imbalances — directly from the electricity paper. No shared mode. Ionic lattice energy is elastic energy of the rope network under ionic tension gradients.
5. Covalent bonding arises from the overlap of rope modes on adjacent atoms, creating bonding and antibonding modes. Bond energy is the energy lowering of the bonding mode. Bond length is the equilibrium overlap distance. Multiple bonds arise from multiple overlapping mode pairs (σ from head-on, π from side-on).

6. Molecular geometry follows from rope mode hybridisation: sp^3 hybrid modes at 109.5° (methane, water), sp^2 at 120° (ethylene, benzene), sp at 180° (CO_2, acetylene). Bond angles are determined by mode geometry, not rope flexibility.
7. Electronegativity is nuclear charge Z as rope tension: higher Z pulls shared bonding modes asymmetrically, producing polar bonds. The bond polarity spectrum from nonpolar covalent to ionic follows from the magnitude of the electronegativity difference between bonding atoms.
8. Pauli exclusion is proposed to arise from rope spin mode saturation: each mode holds at most two kinks of opposite rope-spin ($\pm\hbar/2$), with a third excluded by the standing wave constraint. This is the most important open theoretical problem in rope chemistry; a full derivation requires showing that the rope mode wave equation has no valid three-kink solution within a single mode.

The honest assessment: rope chemistry is quantum chemistry given a physical medium. The Schrödinger equation becomes the rope mode equation; wavefunctions become standing wave patterns; electron probability densities become rope mode amplitudes; quantum superpositions become rope mode mixtures. The mathematics is identical. The physical picture is richer. And the honest boundary beneath the assessment: the Schrödinger equation itself is adopted, not derived — rope chemistry supplies the medium and the mechanism, quantum mechanics supplies the equation, and the corpus keeps that ledger explicit (Section 3.1a).

The rope model does not currently make new quantitative predictions in chemistry beyond what quantum chemistry already provides. Its value is in physical understanding: atomic orbitals are real standing wave patterns in a real physical medium, not abstract probability amplitudes; covalent bonds are physically shared rope modes, not ‘electron pair sharing’ with no mechanism; the periodic table follows from standing wave quantisation plus a maximum of two kinks per mode, not from an unexplained antisymmetry postulate.

The path from rope chemistry to new practical applications runs through the Pauli exclusion derivation. If rope spin mode saturation can be rigorously derived from first principles of the rope network, it would provide a topological foundation for the periodic table and potentially suggest topological approaches to materials design that are not accessible from the standard quantum chemical picture.

Appendix A. The Hydrogen Atom Rope Mode Solution

We summarise the solution to equation (1) in spherical coordinates (r, θ, φ), demonstrating that the rope mode structure exactly reproduces standard quantum chemical results.

The rope wave ψ(r, θ, φ) separates into radial and angular parts:

$\psi_{nlm}(r, \theta, \varphi) = R_{nl}(r) \times Y_{lm}(\theta, \varphi)$ (A1)

where $R_{nl}(r)$ is the radial mode function and $Y_{lm}(\theta, \varphi)$ are the spherical harmonics (the angular mode shapes). The quantum numbers are:

- n = 1, 2, 3, ... (principal mode number; determines energy)
- l = 0, 1, ..., n−1 (angular mode number; s=0, p=1, d=2, f=3)
- m = −l, ..., 0, ..., +l (orientation mode number)

The radial mode functions for hydrogen (Z=1):

$R_{10}(r) = 2(a_0)^{-3/2} \exp(-r/a_0)$ (1s mode) (A2)

$R_{20}(r) = (2a_0)^{-3/2} (1 - r/2a_0) \exp(-r/2a_0)$ (2s mode) (A3)

$R_{21}(r) = (2a_0)^{-3/2} (r/2a_0\sqrt{3}) \exp(-r/2a_0)$ (2p mode) (A4) where a_0 = 0.529 Å is the Bohr radius — the most probable distance for the 1s mode. These are exactly the standard hydrogen atomic wavefunctions. The rope mode picture gives them a physical interpretation: they are standing wave amplitudes of the rope network, not abstract probability amplitudes.

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The Bond Mechanism, Quantitatively

The mode-overlap functional constructed from the network primitives (claims EM-RECON-006/007) makes this paper’s qualitative bond account quantitative: the coupling at 2.5 Å comes out at −0.27 eV once the single stiffness T is fixed by the one-atom 13.6 eV energy (nothing is inserted at the bond scale), and head-on overlap exceeds side-on at every separation, giving the $\sigma > \pi$ ordering (ratio ≈ 4).

Correction to this paper’s earlier language: the repulsive wall at short range is not electrostatic (inter-nuclear Coulomb) in the rope account. It is the strand-extensibility core — the super-quadratic elastic response $c_4 = (k-T_0)/8$, whose positive sign is required by network stability (EM-RECON-009) — the same term that sets nucleon spacing in the nuclear sector. The equilibrium bond length follows as $d_0/\xi = \ln(2b/a)$.

A parameter-free check: the double-exponential potential implies $E''(d_0) = 2|E\_bind|/\xi^2$ exactly (the extensibility coefficient cancels), predicting the H_2 vibrational quantum at 0.634 eV against the measured 0.546 eV — 16% with zero free parameters, and non-circular, since ξ was calibrated to the hydrogen atom, not to the vibration (EM-RECON-010). Chemistry also now exports a cross-sector prediction: the bond geometry extracts k/T_0 ($b/a \approx 2.66$), predicting the nuclear spacing-to-range ratio within 23% with no nuclear input. Core survival under longitudinal relaxation rests on the volume-conservation postulate P-VOL (EM-RECON-013).

Molecular Geometry from First Principles

The hybridisation account of Section 7 imports the quantum-chemical mixing story into rope language: it is the paper's stated equivalence layer, and its mixing coefficients are asserted rather than derived. A dedicated session (registry claim CHEM-GEO-001) has now produced the mechanics-first layer beneath it, with pre-committed pass/fail bars fixed before computation. Three results. FIRST, the heteronuclear bond: the derived mode-overlap cross-term extends to unequal knots exactly analytically — the two-Yukawa convolution $S_{AB}(d) = 4\pi[e^{(-d/\xi_A)} - e^{(-d/\xi_B)}]/((1/\xi_B^2 - 1/\xi_A^2)d)$, recovering the homonuclear limit — and its contact-core-plus-slow-tail structure makes covalent-radius ADDITIVITY a derived form rather than an empirical rule. Blind test from homonuclear anchors alone: H–Cl predicted 1.365 Å against 1.275 measured (7.0% high); O–H 1.108 against 0.958 (15.7% high) — inside the parameter-free tier this paper's H_2 vibration check established, with both deviations of the SAME sign: polar bonds run shorter than additive (the Schomaker–Stevenson contraction), registered as the identified next-order effect (asymmetric tail penetration), not tuned away. SECOND, two angular theorems, verified by direct integration: (i) PHASE-BLOCKING — a dipolar mode's two lobes carry opposite phase, so partners attached at opposite lobes acquire overlaps of opposite sign, one bonding and one antibonding: a single dipolar mode cannot host two σ bonds, which forbids linear di-attachment at leading order; (ii) ORTHOGONALITY — distinct dipolar modes are exactly orthogonal, so the second bond takes a perpendicular lobe: two single bonds on one centre sit at 90° at leading order, opened upward by the nonbonded contact repulsion of the attached partners. THIRD, the pre-committed bar — discriminate bent H_2O from linear CO_2 before any angle is requested — is MET: H_2O bent (derived), CO_2 collinear via the π-orthogonality constraints of its two double bonds (Modeled, pending the hybrid-mode derivation).
RECONCILIATION WITH SECTION 7, stated honestly. The raw-lobe theorems give 90°-plus-opening; full sp^3 mixing gives 109.5°-minus-lone-pair-compression; water's measured 104.5° lies between the two brackets, and which bracket dominates is precisely the question the pending hybridisation derivation must answer from rope mechanics. The periodic trend supports the layered picture: in H_2S, where the heavier centre hybridises weakly, the measured angle is 92.1° — almost exactly the raw orthogonal-lobe prediction — while water, the strongest hybridiser of its column, opens furthest toward the sp^3 limit. The mechanics-first layer (Derived) and the equivalence layer (Modeled: asserted mixing) are thus complementary accounts at different orders, and the registry carries each at its own strength. Benchmark: benchmarks/em/molecular_geometry_foundations.py.

Reaction Dynamics: Why Reactions Happen

Chemistry is not only equilibrium; the question this section answers, with pre-committed pass/fail bars fixed before computation, is why reactions have barriers at all. THE MECHANICAL ANSWER: because one mode cannot fully bond two partners. In a collinear exchange A–B–C the central atom's single bonding mode is subject to the same capacity mechanics that shaped molecular geometry, and three-center sharing is strictly sub-additive: the bonding level $-\sqrt{(t_1^2 + t_2^2)}$ is always less binding than two full bonds. The exchange path therefore passes through configurations of partial frustration, and the sub-additivity forces the end atoms into contact range before the transfer completes. Building the three-center surface from this paper's own Morse-consistent diatomic (attraction De e^{-x}, core De e^{-2x}, ξ from the H_2 vibration relation), the coherent sharing limit yields a fully analytic result: the frustration term bottoms at exactly −1 De at $x^* = \ln\sqrt{2}$, so the entire coherent barrier is the third-body contact,

$(1/4)e^{(-2d_0/\xi)}$ De = 0.0142 De. Barrier existence is thereby DERIVED, in both sharing limits, from two named mechanisms — frustration and third-body contact — with neither assumed. THE QUANTITATIVE VERDICT, stated as the honest bracket it is: the two derivable sharing rules — coherent eigenvalue sharing (barrier 0.0142 De, 0.067 eV) and incoherent capacity partition (0.50 De, 2.37 eV) — BRACKET the measured H + H_2 barrier ratio (0.42/4.75 = 0.089) parameter-free. An observation recorded but deliberately not claimed: the bracket's logarithmic midpoint is 0.084, within 5% of the measured value; the physical system is partially coherent, and deriving the coherence fraction of two-strand mode transfer — how much phase memory survives the crossing — is the registered next step that would close the bracket to a point prediction, or expose the midpoint as coincidence. Two classics of physical chemistry then emerge from the surface unbidden. HAMMOND'S POSTULATE: on the heteronuclear surface the saddle sits with the forming strong bond still long and the breaking weak bond barely stretched — an early transition state for the exothermic direction, derived rather than asserted. CATALYSIS acquires its mechanical reading: a catalyst DONATES MODE CAPACITY at the crossing, relaxing the frustration constraint (in the toy model the barrier falls monotonically from 0.875 to 0.500 De as donated capacity grows, with the catalyst returned unchanged) — the rope-mechanical content of ‘catalysts provide alternative pathways.’ Statuses per the registry: barrier existence and the Hammond direction Derived in-model; the bracket Derived at both ends with the interior open; catalysis Modeled. Benchmark: benchmarks/em/reaction_dynamics_foundations.py.
FIELD-LEVEL CONFIRMATION (CHEM-DYN-002). The barrier results above were first derived with exponential-hopping stand-in amplitudes; the frozen mode-overlap functional has now been evaluated along the same exchange coordinate, with the amplitude shape computed from grid integrals of the declared profiles and everything else inherited unchanged. The barrier survives: +0.0122 De coherent (vs the ansatz's 0.0142; the field shape's heavier tail slightly lowers it), saddle exactly symmetric, incoherent limit 0.502, the parameter-free bracket still containing the measured 0.0885 (log-midpoint 0.078, recorded as observation only), and Hammond's early transition state emerging from computed amplitudes. Barrier existence is thus a property of the derived functional, not the stand-in. The coherence fraction remains the registered next step; rates and tunneling belong to the quantum-kinetic layer. Benchmark: benchmarks/em/reaction_dynamics_field_level.py.
CATALYSIS, HAMMOND, AND BELL–EVANS–POLANYI (CHEM-DYN-003). The field-level program closes with three structural results. The catalysis shape-independence is now exact and derived: the capacity-donation barrier depends only on the reachable amplitude range, so field amplitudes reproduce the analytic $1-(1+\kappa)^2/8$ to four decimals at every κ. Hammond's postulate emerges as a THRESHOLD: the thermoneutral saddle is symmetric, and any appreciable exothermicity pushes the transition state maximally early. And the Bell–Evans–Polanyi correlation holds with its extreme reported straight: in the coherent two-channel surface, exothermic steps are barrierless — the barrier belongs to the thermoneutral exchange — with the reverse barrier tracking endothermicity exactly. The honest caveat is stated in the claim: this surface lacks the entrance-channel contact structure that gives real exothermic reactions their small barriers; the model captures the direction and the extreme, not asymmetric barrier magnitudes. Benchmark: benchmarks/em/reaction_dynamics_catalysis.py.
THE ENTRANCE CHANNEL (CHEM-DYN-004). The caveat above was answered by the hydrogen-bond arc's own machinery: the incoming atom's contact with the old bond's occupied pair mode -- the capacity-saturated $+t^2$ penalty, gated by the eigenvector occupancy weights, zero

new constants. The parameter-free headline: the coherent thermoneutral barrier rises from 0.0122 to 0.0791 De, eleven percent from the measured $H+H_2$ ratio 0.0885 -- most of what the bracket attributed to an unknown coherence fraction was derivable contact structure. Hammond softens from threshold to the textbook gradient; mild exothermicity gains a genuine small barrier while strong exothermicity remains barrierless -- the literal all-positive bar failed and is kept, with the observation that the restored pattern is the empirically real one. Benchmark: benchmarks/em/reaction_entrance_channel.py.

THE BRACKET RESTATED AND THE FIRST COHERENCE NUMBER (CHEM-DYN-005). Porting the entrance term to the incoherent surface moves its edge only slightly (0.502 → 0.525 — the max rule's shape-insensitivity extends to entrance-insensitivity), so the barrier bracket restates as [0.0791, 0.525] with the measured 0.0885 sitting just above the coherent edge. That geometry yields the program's first coherence number: an inferred incoherent admixture of 2.1 percent — the $H+H_2$ crossing runs ~98 percent coherent. This is an inference from where data falls between derived edges, not a derivation; but the open problem now has a sharp target: derive $f \approx 0.02$. Catalysis survives the entrance correction intact — a donated third channel, present throughout as the physical pre-complex, lowers the barrier monotonically with the catalyst returned to machine precision. Benchmark: benchmarks/em/bracket_and_catalysis_entrance.py.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **CHEM-DYN-001 [Modeled]:** OPEN [benchmark: benchmarks/em/reaction_dynamics_foundations.py]
- **CHEM-DYN-002 [Modeled]: Field-level barrier confirmation -- the ansatz survives the derived functional [benchmark: benchmarks/em/reaction_dynamics_field_level.py]**
- **CHEM-DYN-003 [Modeled]: Catalysis invariance exact; Hammond threshold; BEP extreme [benchmark: benchmarks/em/reaction_dynamics_catalysis.py]**
- **CHEM-DYN-004 [Modeled]: Entrance channel -- coherent barrier to 11% of measured; Hammond gradient; bar B1 partial, kept [benchmark: benchmarks/em/reaction_entrance_channel.py]**
- **CHEM-DYN-005 [Modeled]: Bracket restated [0.0791, 0.525]; inferred coherence ~98%; catalysis intact on the corrected surface [benchmark: benchmarks/em/bracket_and_catalysis_entrance.py]**
- **CHEM-GEO-001 [Modeled]:** OPEN [benchmark: benchmarks/em/molecular_geometry_foundations.py]
- **CHEM-HB-001 [Modeled]:** HYDROGEN BONDING COMPLETED [benchmark: benchmarks/em/hydrogen_bond_electrostatics.py]
- **CHEM-HB-002 [Modeled]:** HYDROGEN-BOND ORDERING TEST [benchmark: benchmarks/em/hydrogen_bond_ordering.py]
- **CHEM-HB-003 [Modeled]:** LONE-PAIR DIPOLE FROM OCCUPIED LOBES [benchmark: benchmarks/em/lone_pair_dipole_derivation.py]
- **CHEM-HB-004 [Modeled]:** CONTACT-REPULSION TEST [benchmark: benchmarks/em/contact_repulsion_test.py]

- **CHEM-HB-005 [Modeled]: TAIL-ASYMMETRY CHARGES -- derived map repairs O and N; F route-convergent, bar failed 2-of-3 [benchmark: benchmarks/em/tail_asymmetry_charges.py]**
- **CHEM-HB-006 [Modeled]: CONTACT ANISOTROPY -- O to 1.5%, N to 15%; F terminal (half the required anisotropy) [benchmark: benchmarks/em/contact_anisotropy.py]**
- **CHEM-MET-001 [Modeled; 2.8x shortfall decomposed 5 Sep 2026, MET-DISCRIM-FAIL]:** OPEN [benchmark: benchmarks/em/metallic_bonding_foundations.py]
- **CHEM-STRUCT-001 [Derived]:** Periodic shell capacities 2n^2 from n^2 rope modes x two spin states [benchmark: benchmarks/em/periodic_shell_counting.py]